Experiment 06

Study of Transmission Media

Lab Objective:

To study and understand different types of **transmission media** used in computer networking, including their **characteristics, advantages, disadvantages**, and **usage scenarios**.

Prerequisites:

- Basic knowledge of **networking concepts** (like LAN, WAN, topology).
- Understanding of **data transmission** principles (analog/digital, bandwidth, signal attenuation).

Outcome:

By the end of this experiment, students will be able to:

- Identify different types of **transmission media**.
- Describe their **working principles**.
- Compare **advantages, limitations**, and **usage scenarios** of each.

Transmission Media in Networking

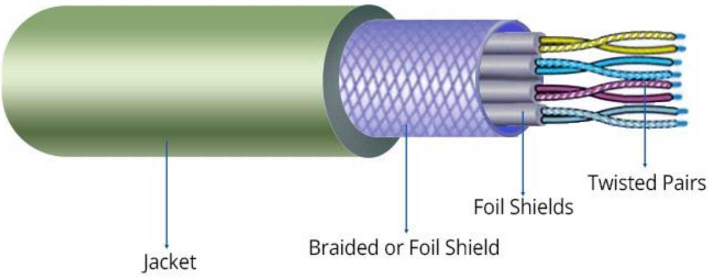
Transmission media refers to the physical or wireless pathway through which data is transmitted from one device to another in a network. It is broadly classified into two main types:

1. Guided Media (Wired Transmission)

Data signals are guided through a physical medium such as cables.

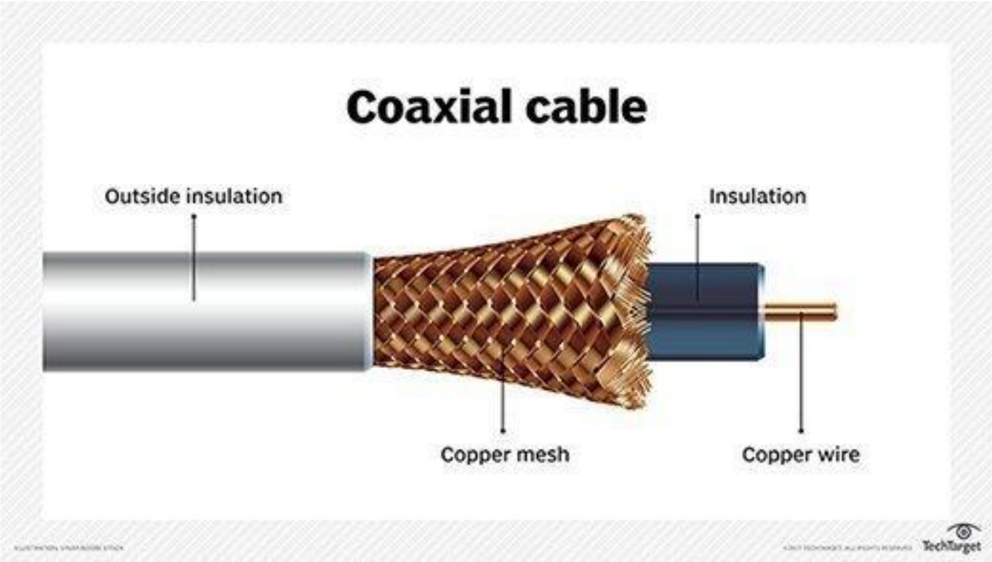
a) Twisted Pair Cable

- **Description:** Consists of pairs of insulated copper wires twisted together.
- **Types:**
    - **UTP (Unshielded Twisted Pair)** – Used in LANs.
    - **STP (Shielded Twisted Pair)** – Provides better noise resistance.
- **Characteristics:**
    - Speed: Up to 10 Gbps (Cat6 or higher).
    - Max Length: ~100 meters.
    - Cost: Low.
- **Usage:** Telephone lines, LAN cables (Ethernet).
- **Advantages:** Cheap, flexible, easy to install.
- **Disadvantages:** Susceptible to EMI and crosstalk.



b) Coaxial Cable

- **Description:** Copper core surrounded by insulation and shielding.
- **Characteristics:**
    - Speed: Moderate (10-100 Mbps).
    - Distance: Up to 500 meters.
    - Cost: Moderate.
- **Usage:** Cable TV, older Ethernet networks.
- **Advantages:** Better noise resistance than twisted pair.
- **Disadvantages:** Bulkier and more expensive.



c) Optical Fiber Cable

- **Description:** Uses light signals to transmit data via glass or plastic fibers.
- **Types:**
  - **Single-mode fiber** – Long distance.
  - **Multi-mode fiber** – Short distance.

- **Characteristics:**
  - Speed: Extremely high (Gbps to Tbps).
  - Distance: Several kilometers.
  - Cost: High.
- **Usage:** Internet backbone, ISPs, data centers.
- **Advantages:** Immune to EMI, high bandwidth.
- **Disadvantages:** Expensive, difficult to install/repair.

2. Unguided Media (Wireless Transmission)

Uses the air or vacuum to transmit signals.

a) Radio Waves

- **Description:** Electromagnetic waves with long range and good penetration.
- **Frequency:** 3 kHz – 1 GHz.
- **Usage:** AM/FM radio, mobile phones, wireless LANs.
- **Advantages:** Long range, no cables needed.
- **Disadvantages:** Susceptible to interference.

b) Microwaves

- **Description:** Travel in straight lines, require line-of-sight.
- **Frequency:** 1 GHz – 300 GHz.
- **Usage:** Satellite communication, mobile networks.
- **Advantages:** High bandwidth.
- **Disadvantages:** Affected by weather, needs alignment.

c) Infrared

- **Description:** Short-range wireless communication using light signals.
- **Usage:** Remote controls, short-distance file transfer (IR).
- **Advantages:** Secure in closed rooms.
- **Disadvantages:** Cannot penetrate walls.

d) Satellite Communication

- **Description:** Uses satellites to relay signals over long distances.
- **Usage:** Global internet access, GPS, DTH.
- **Advantages:** Very long-distance coverage.
- **Disadvantages:** Expensive, weather-dependent, high latency.

Comparison Table:

Media Type	Speed	Cost	Distance	EMI Resistance	Common Use
Twisted Pair	Up to 10 Gbps	Low	~100m	Low	LAN, telephone
Coaxial Cable	Moderate	Medium	~500m	Medium	Cable TV
Optical Fiber	Very High	High	>10 km	High	Internet backbone
Radio Waves	Moderate	Low	Long	Low	Wi-Fi, radio, cell phones
Microwaves	High	Medium	Long (LoS)	Medium	Satellites, microwave links
Infrared	Low	Low	Very Short	High	Remote controls
Satellite	Moderate	Very High	Global	High	GPS, DTH, global internet

Conclusion:

Transmission media plays a vital role in determining the **speed, range, cost**, and **reliability** of a network. Choosing the right media depends on:

- **Distance**
- **Data speed requirement**
- **Budget**
- **Environment (indoor/outdoor)**